

What is Data Definition Language-55

Introduction

In SQL, commands are divided into categories based on their purpose. One of the most important categories is **Data Definition Language (DDL)**. DDL statements are used to **define, create, modify, and delete database structures** like tables, schemas, indexes, and constraints.

DDL does not deal with the actual data stored inside the tables. Instead, it manages the **structure** of the database. In MySQL Workbench, DDL queries are executed in the query editor and once executed, the changes are applied immediately.

Features of DDL

- Defines the database schema (structure).
- Automatically commits changes. Once executed, rollback is not possible.
- Does not manipulate data, only structure.
- Works on objects like tables, views, indexes, schemas, and constraints.

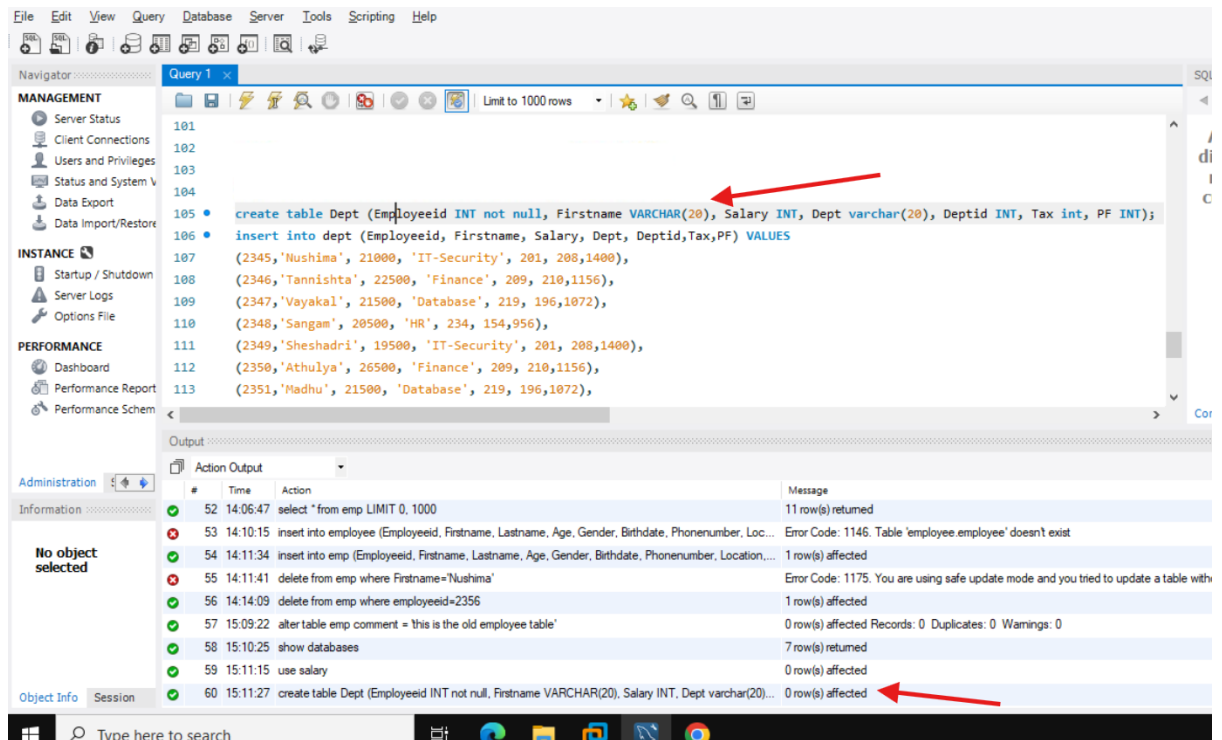
DDL Commands

1. CREATE

Used to create a database or database objects.

- Example: Create a new table Employees.

```
create table Dept (Employeeid INT not null, Firstname VARCHAR(20), Salary INT, Dept  
varchar(20), Deptid INT, Tax int, PF INT);
```



- This defines a new table with specific columns.

2. ALTER

Used to modify an existing table structure.

- Example 1: alter with comment

We can use the comment query using alter

```

91 • use employee;
92 • rename table employee to emp;
93 • select * from emp;
94 • truncate table emp;
95 • update emp set phonenum=909090909 where employeeid=2355;
96 • insert into emp (Employeeid, Firstname, Lastname, Age, Gender, Birthdate, Phonenum, Location, Maritalstatus)
97   VALUES (2356, 'Nushima', 'Noushadh', 21, 'Female', '2004-05-01', 987654321, 'Coimbatore', 'Single');
98 • delete from emp where employeeid=2356;
99 • alter table emp comment = 'this is the old employee table';
100

```

#	Time	Action	Message
46	12:58:16	use employee	0 row(s) affected
47	12:58:20	select * from emp LIMIT 0, 1000	11 row(s) returned
48	14:05:58	update emp set phonenum=909090909 where Employeeid=2355	Error Code: 1264. Out of range value for column 'Phonenum' at row 1
49	14:06:03	select * from emp LIMIT 0, 1000	11 row(s) returned
50	14:06:32	update emp set phonenum=909090909 where employeeid=2355	Error Code: 1264. Out of range value for column 'Phonenum' at row 1
51	14:06:43	update emp set phonenum=909090909 where employeeid=2355	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
52	14:06:47	select * from emp LIMIT 0, 1000	11 row(s) returned
53	14:10:15	insert into employee (Employeeid, Firstname, Lastname, Age, Gender, Birthdate, Phonenum, Loc...	Error Code: 1146. Table 'employee.employee' doesn't exist
54	14:11:34	insert into emp (Employeeid, Firstname, Lastname, Age, Gender, Birthdate, Phonenum, Location,...	1 row(s) affected
55	14:11:41	delete from emp where Firstname='Nushima'	Error Code: 1175. You are using safe update mode and you tried to update a table without a
56	14:14:09	delete from emp where employeeid=2356	1 row(s) affected
57	15:09:22	alter table emp comment = 'this is the old employee table'	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

See the comment as been added to emp table

Query 1: employee.emp

Local instance MySQL80

employee.emp

File format:

Data path: C:\ProgramData\MySQL\MySQL Server 8.0\Data\employee\emp.ibd

Update time: 2025-08-26 14:14:09

Create time: 2025-08-26 15:31:29

Auto increment:

Table collation: utf8mb4_0900_ai_ci

Create options:

Comment: this is the old employee table

Information on this page may be outdated. Click [Analyze Table](#) to update it.

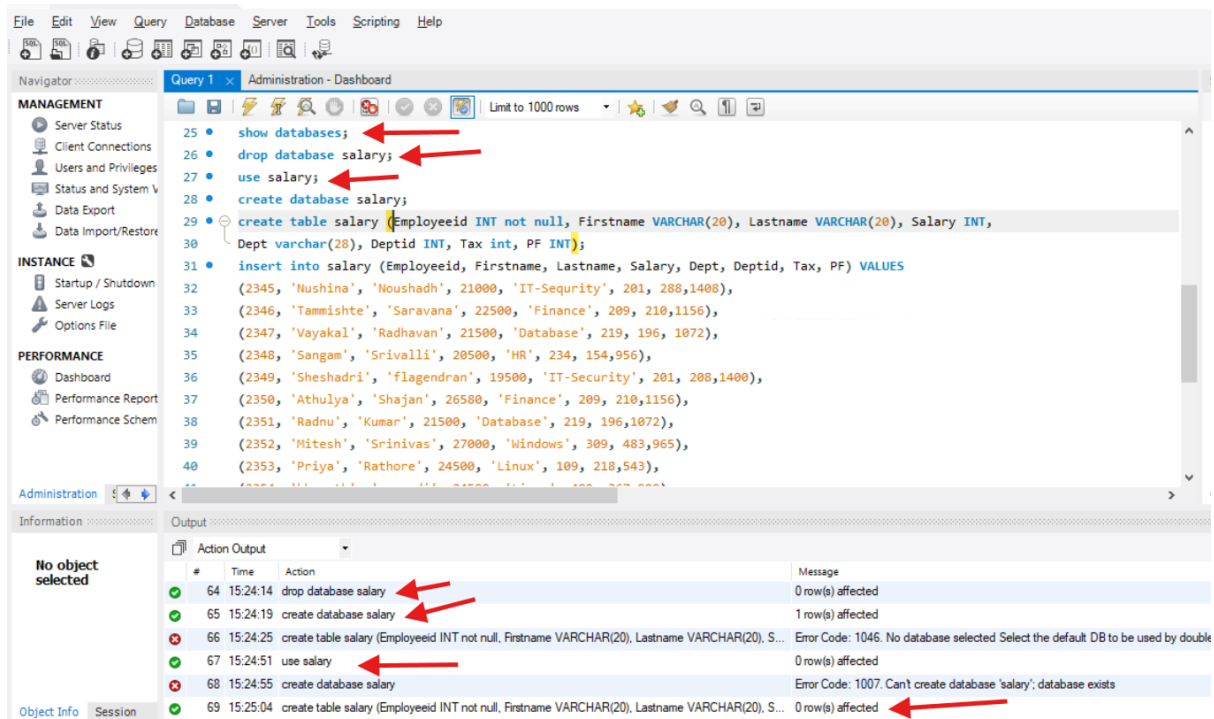
#	Time	Action	Message	Duration / Fetch
66	15:20:45	select salary.Firstname, dept.Dept from salary full join dept on salary.Deptid = dept.Deptid L...	Error Code: 1054. Unknown column 'salary.Firstname' in field list	0.000 sec
67	15:20:56	select salary.Firstname, dept.Dept from salary cross join salary.Dept LIMIT 0, 1000	100 row(s) returned	0.000 sec / 0.000 sec
68	15:23:06	desc emp	Error Code: 1146. Table 'salary.emp' doesn't exist	0.000 sec
69	15:23:21	desc salary	8 row(s) returned	0.000 sec / 0.000 sec
70	15:29:33	ANALYZE TABLE 'employee'.employee	OK	0.000 sec
71	15:30:55	show databases	7 row(s) returned	0.000 sec / 0.000 sec
72	15:31:02	use emp	Error Code: 1049. Unknown database 'emp'	0.000 sec
73	15:31:18	use employee	0 row(s) affected	0.000 sec
74	15:31:29	alter table emp comment = 'this is the old employee table'	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.000 sec

3. DROP

Used to delete a database object permanently.

- Example:

DROP database Salary;



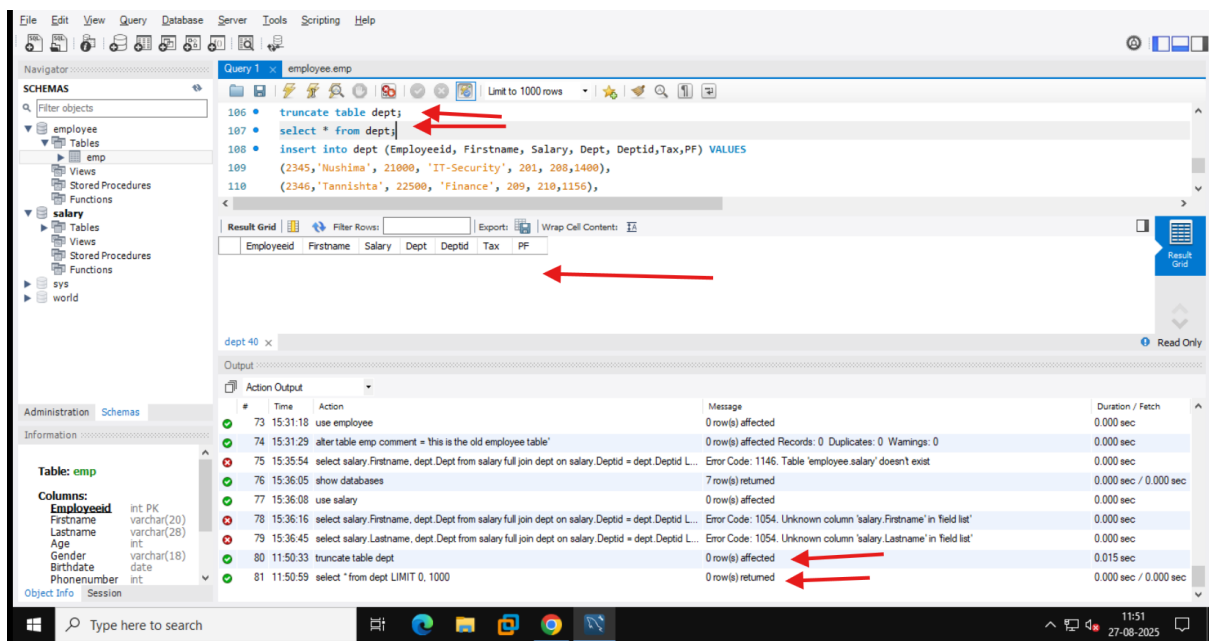
- This removes the tables and tables and all its data.

4. TRUNCATE

Used to remove all rows from a table but keep the structure.

- Example:

TRUNCATE TABLE dept;

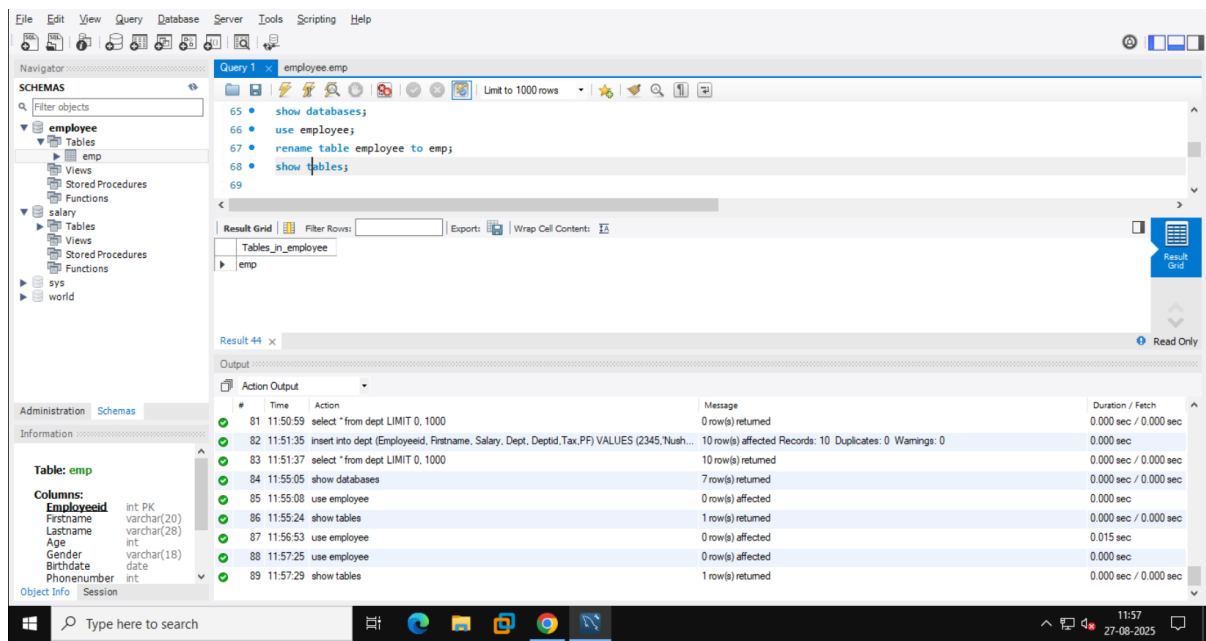


- After execution, the table is empty but still exists.

5. RENAME

Used to rename an existing table.

- Example:
- `RENAME TABLE Employee TO emp;`



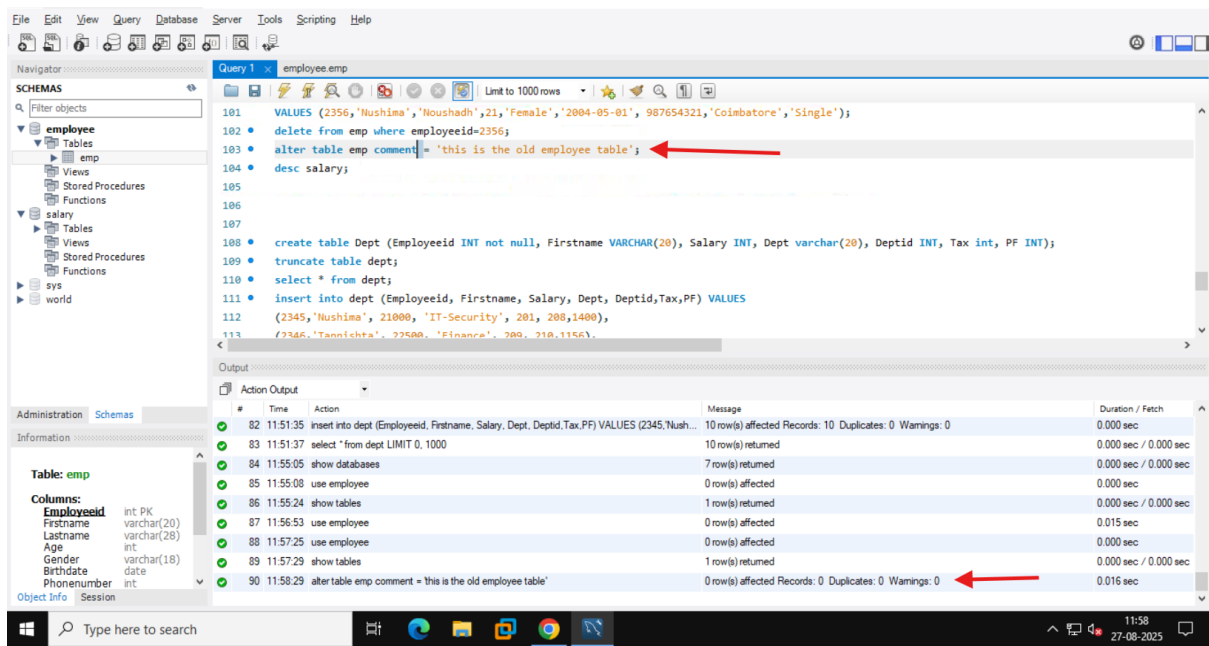
6. COMMENT

Used to add comments on schema objects for documentation.

- Example:

```
CREATE TABLE Dept (  
    DeptID INT PRIMARY KEY,  
    DeptName VARCHAR(100) COMMENT 'Department Name'  
);
```

Here in workbench we have made a comment on emp table using alter



Example Workflow in MySQL Workbench

1. Create a table:

```

CREATE TABLE Dept (
    DeptID INT PRIMARY KEY,
    DeptName VARCHAR(50)
);

```

2. Add a column:

```
ALTER TABLE Dept ADD Location VARCHAR(100);
```

3. Rename the table:

```
RENAME TABLE Dept TO Department;
```

4. Truncate data:

```
TRUNCATE TABLE Department;
```

5. Drop table:

```
DROP TABLE Department;
```

Summary

- DDL defines the structure of the database.
- Key commands: CREATE, ALTER, DROP, TRUNCATE, RENAME, COMMENT.

- Once executed, changes are permanent.
- DDL commands manage **how data is stored**, not the data itself.